

**Figure 1 — Generation Success Rate Comparison**

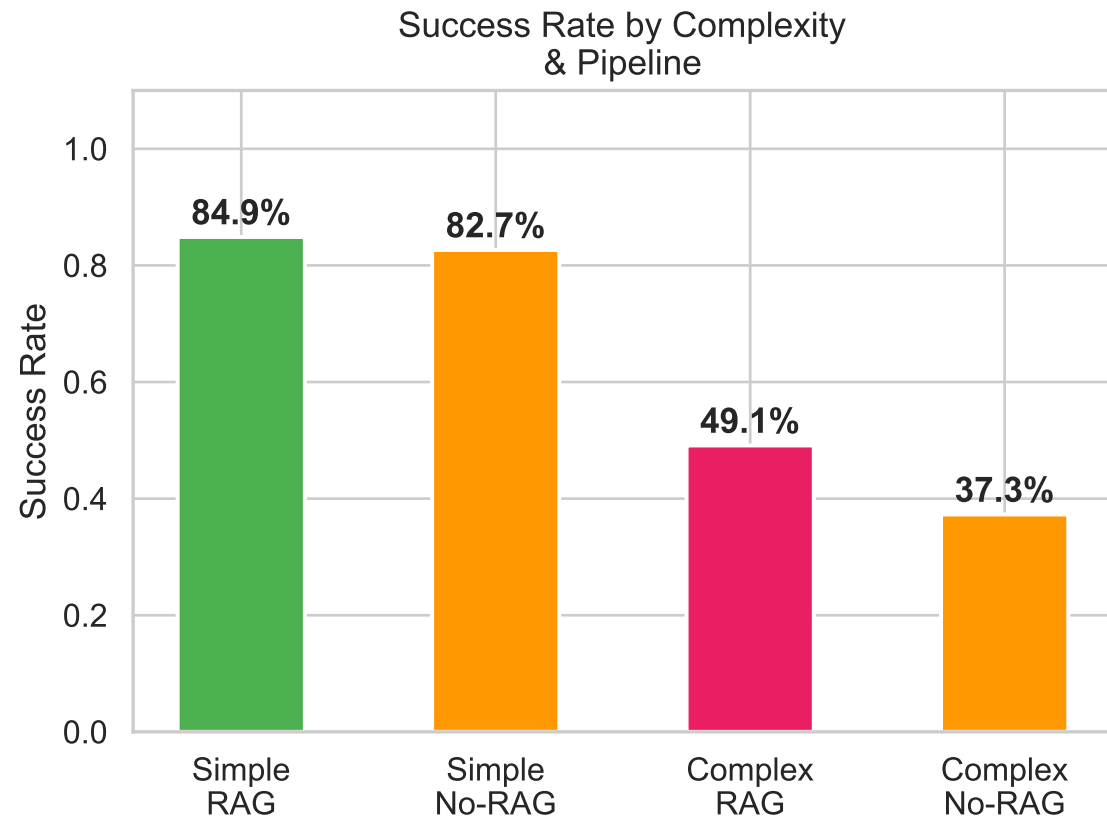
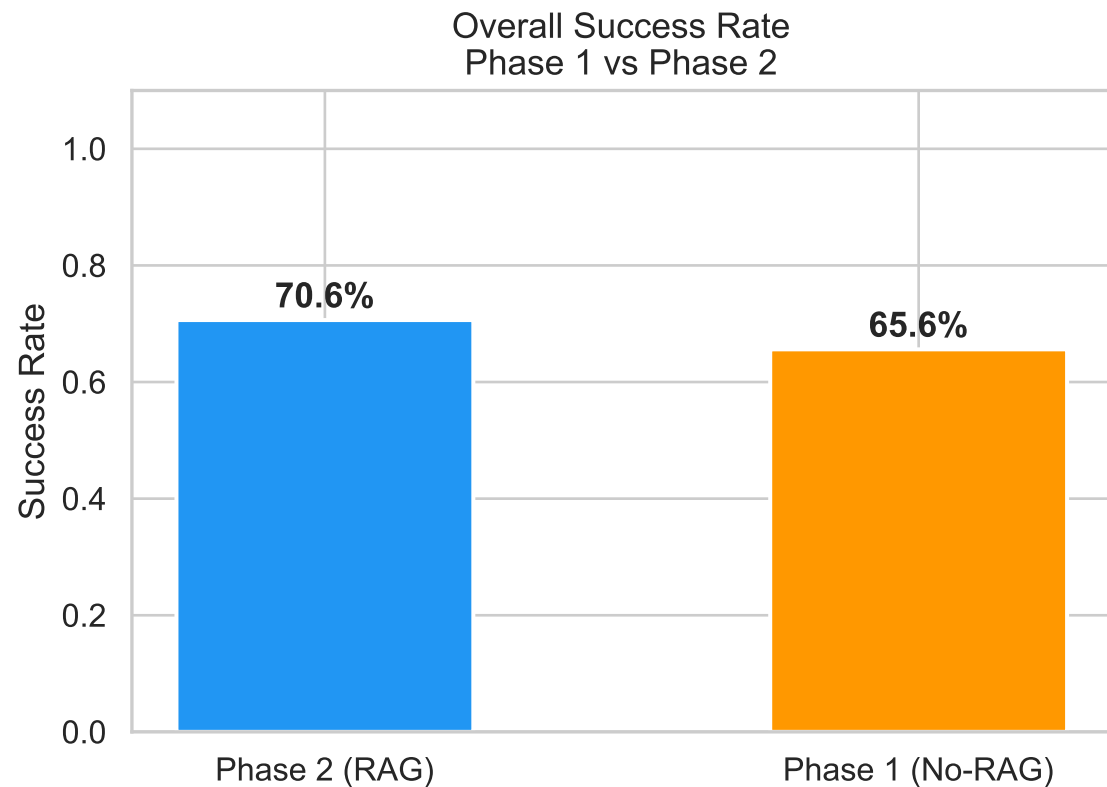
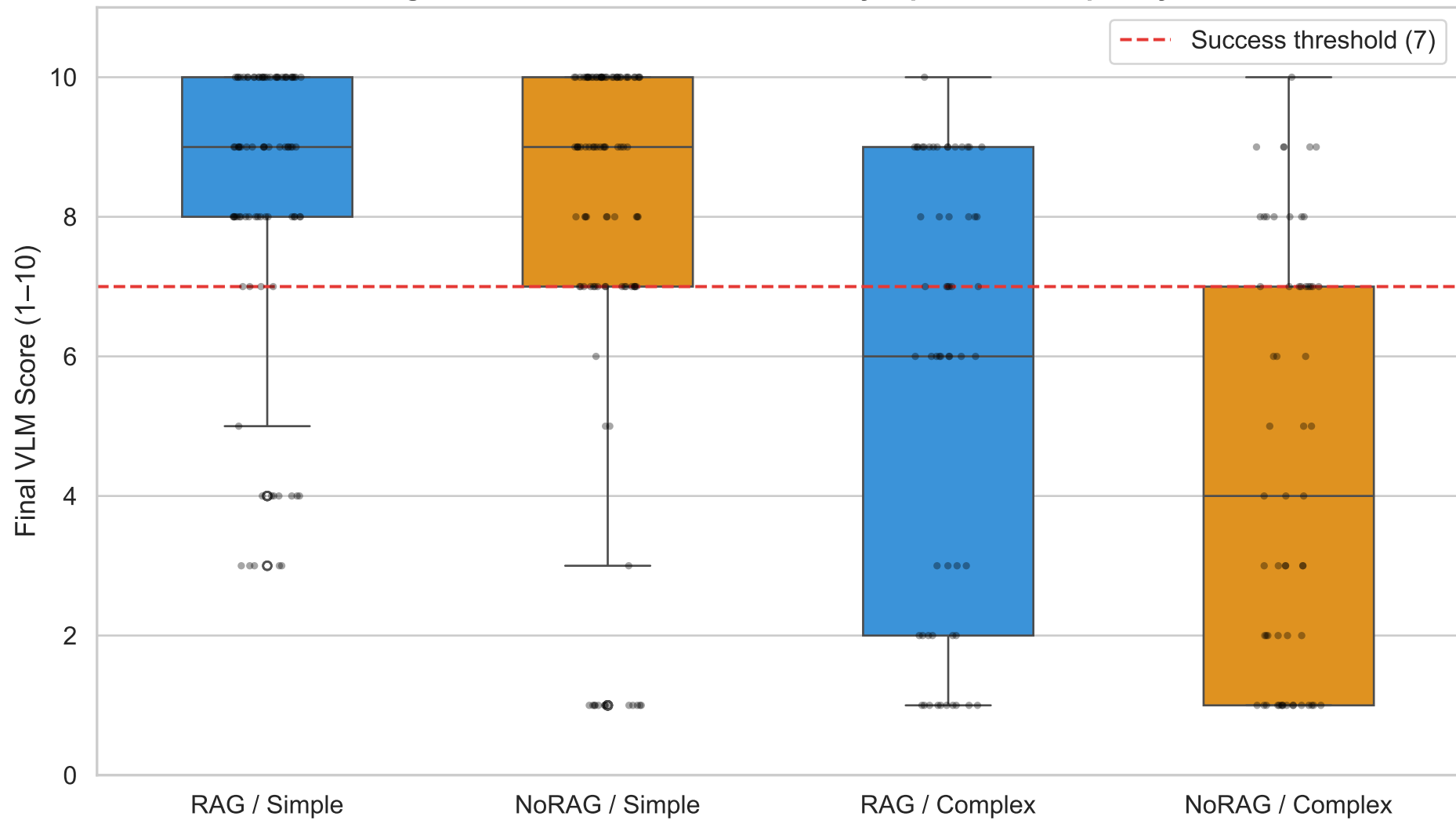


Figure 2 — VLM Score Distribution by Pipeline & Complexity



**Figure 3 — Average Iterations to Reach Success**

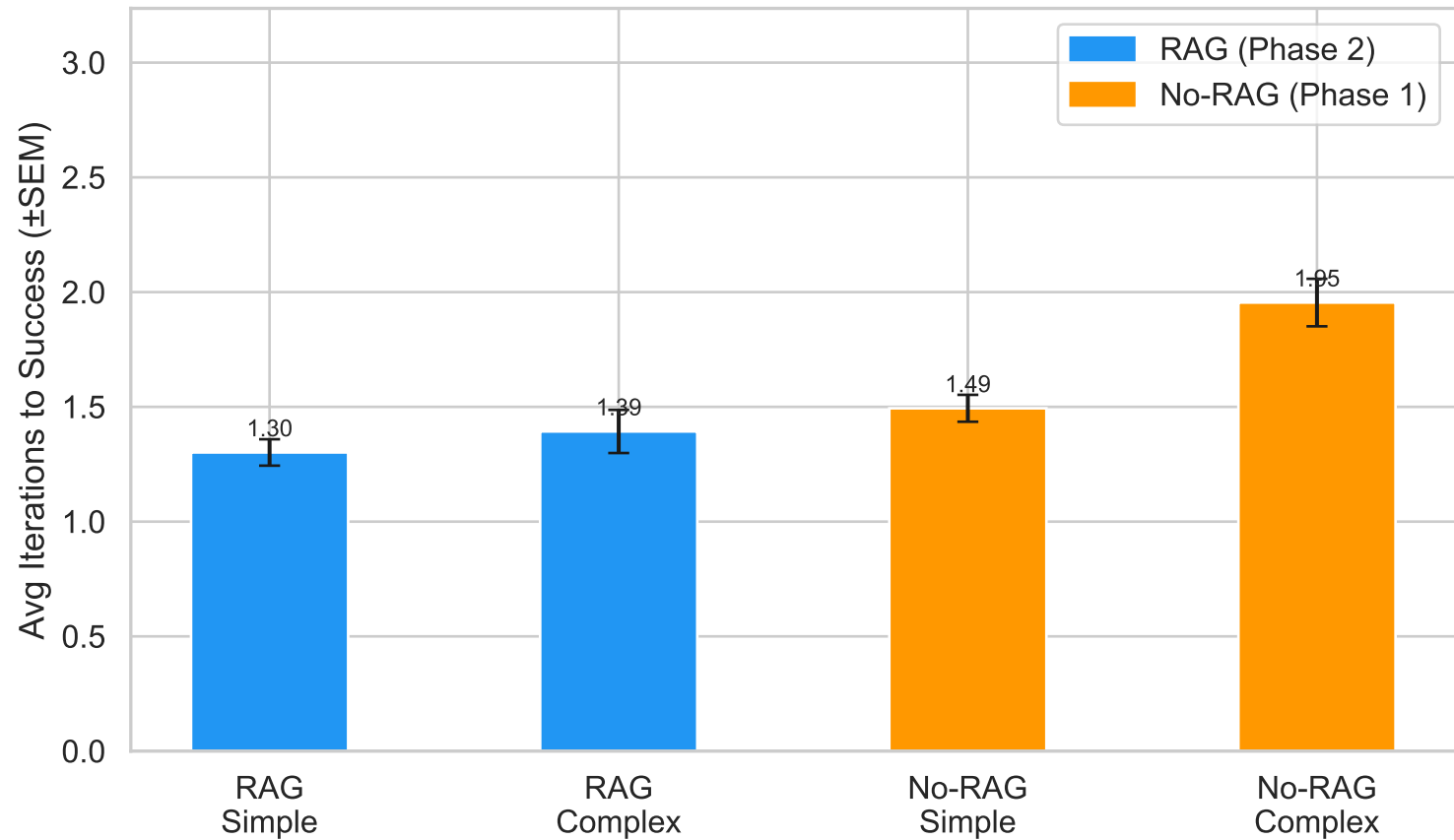
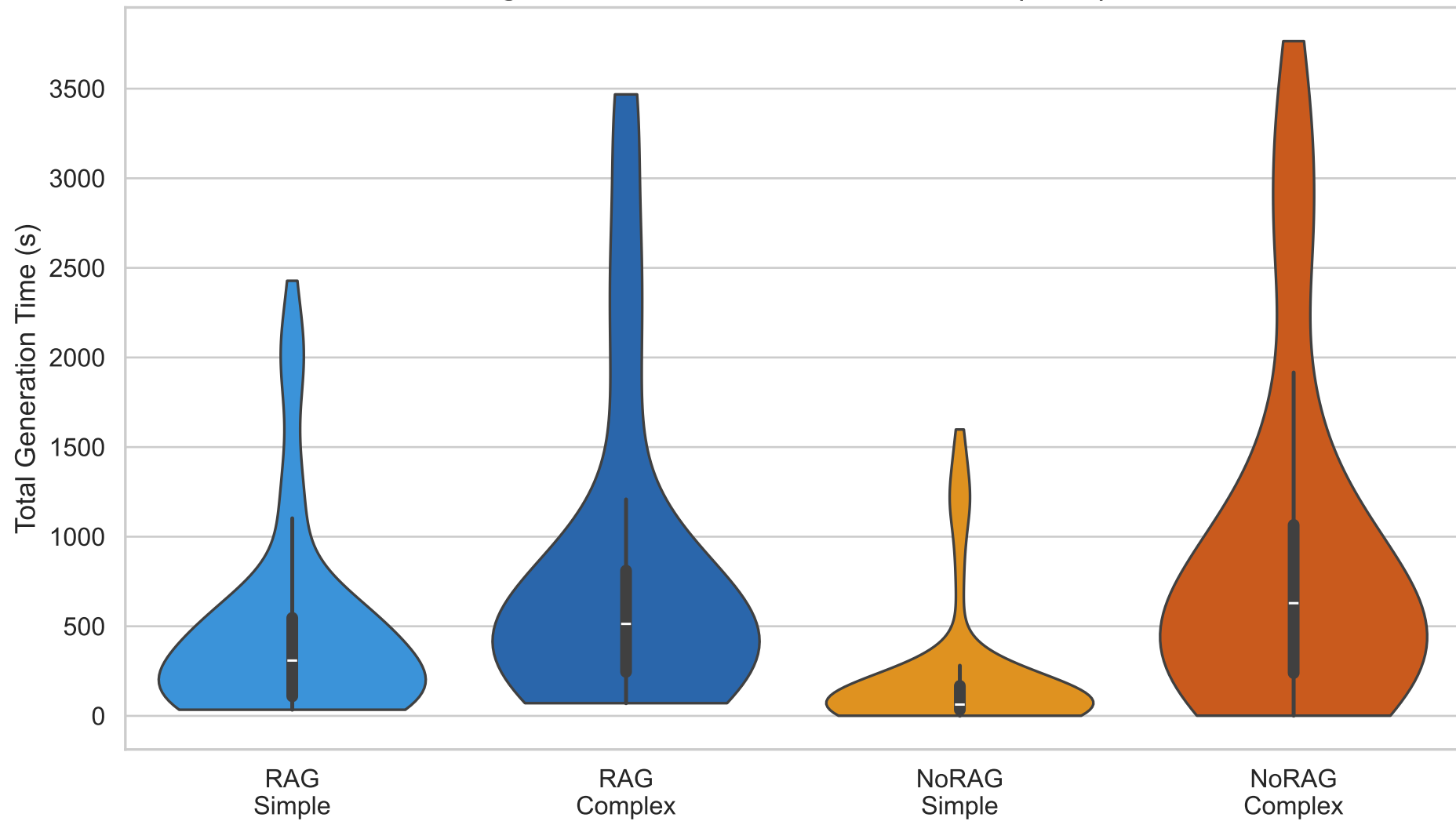


Figure 4 — Generation Time Distribution (Violin)



**Figure 5 — First-Pass HLSL Compile Success Rate**

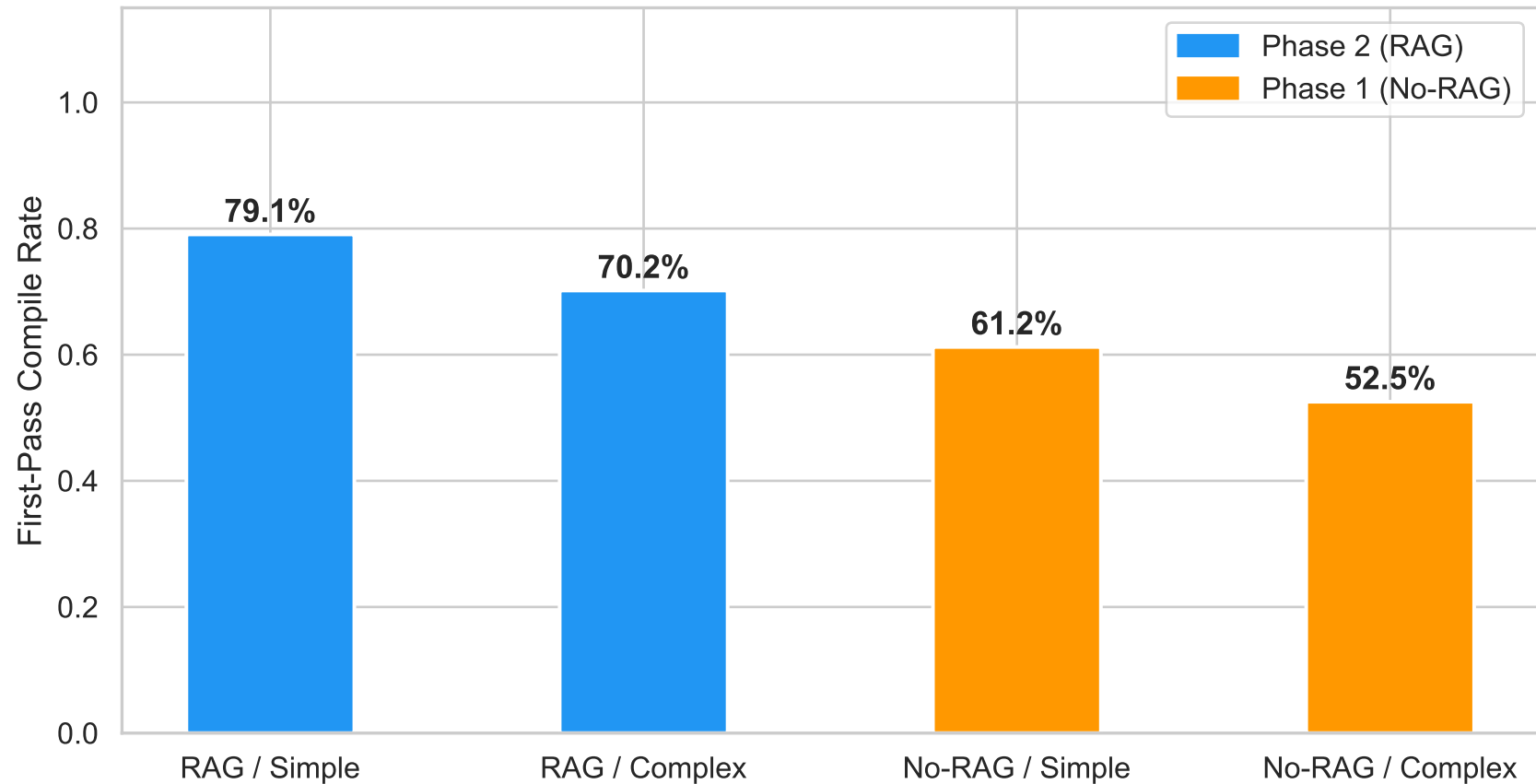
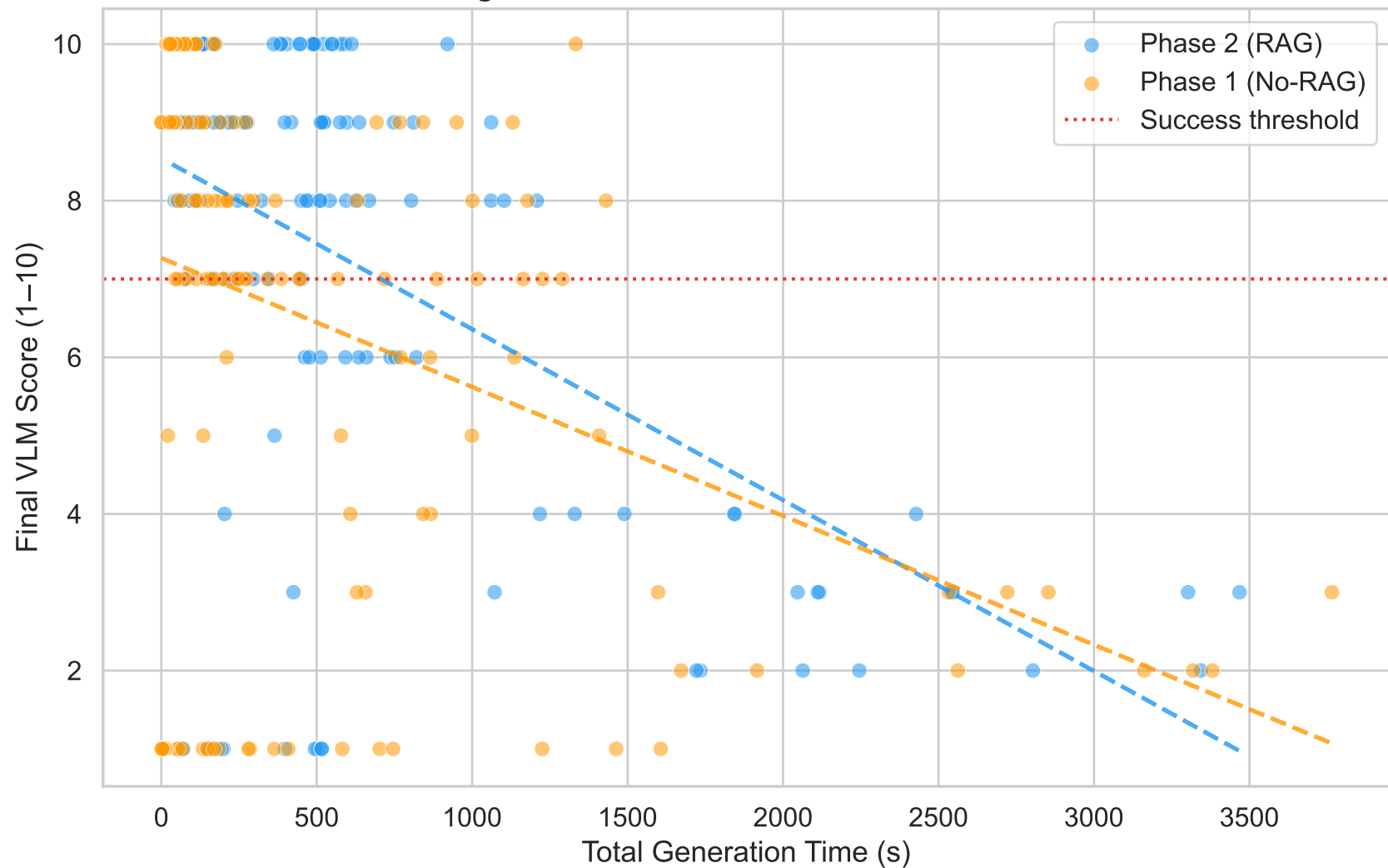
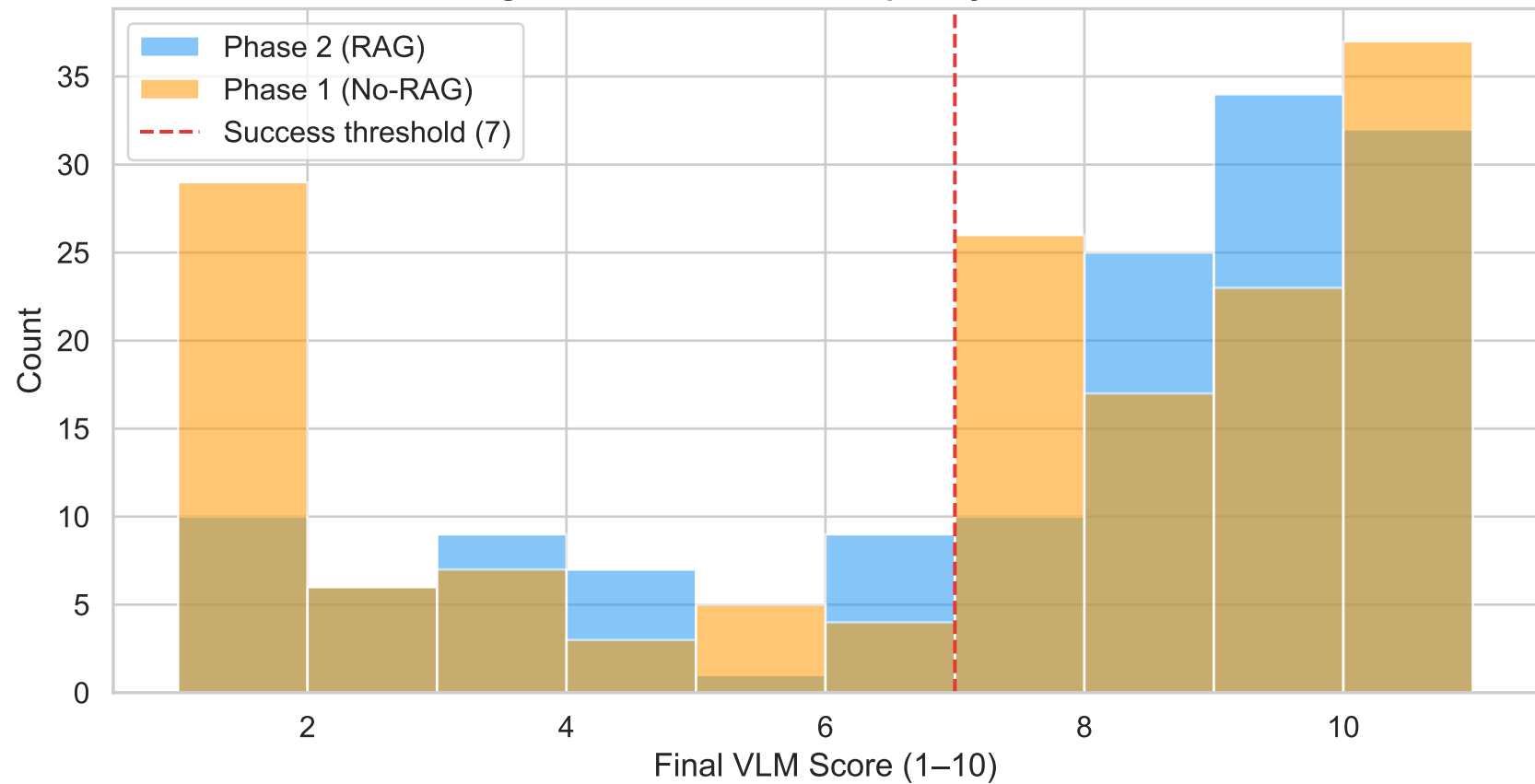


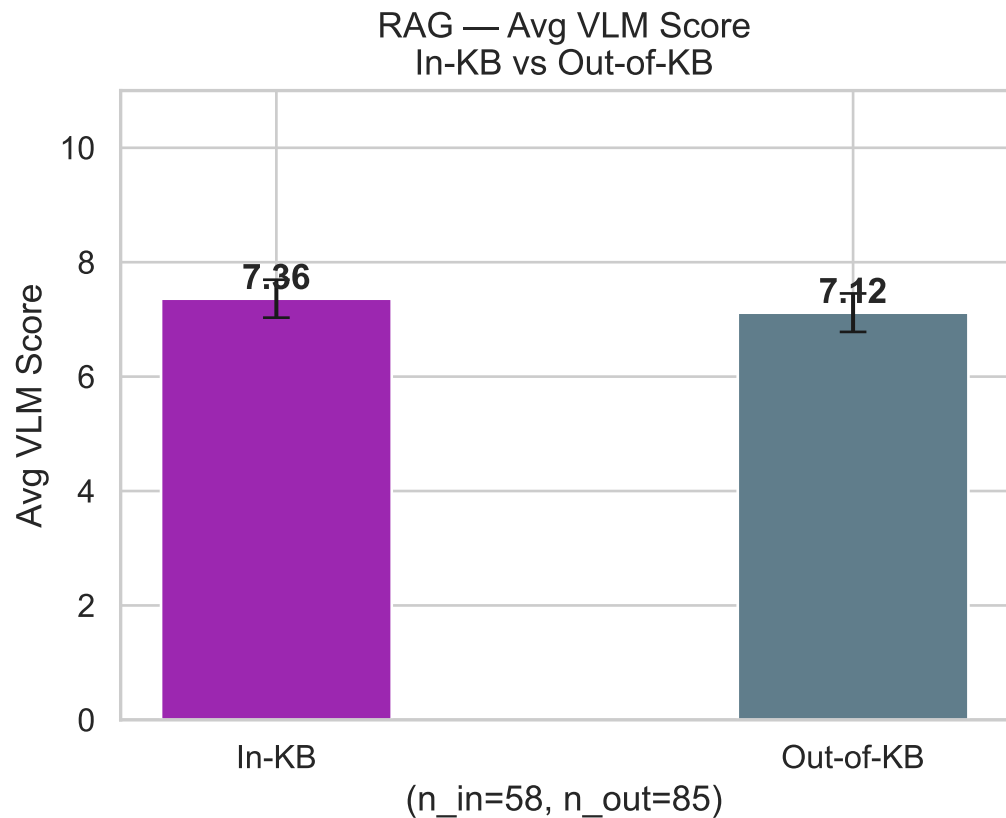
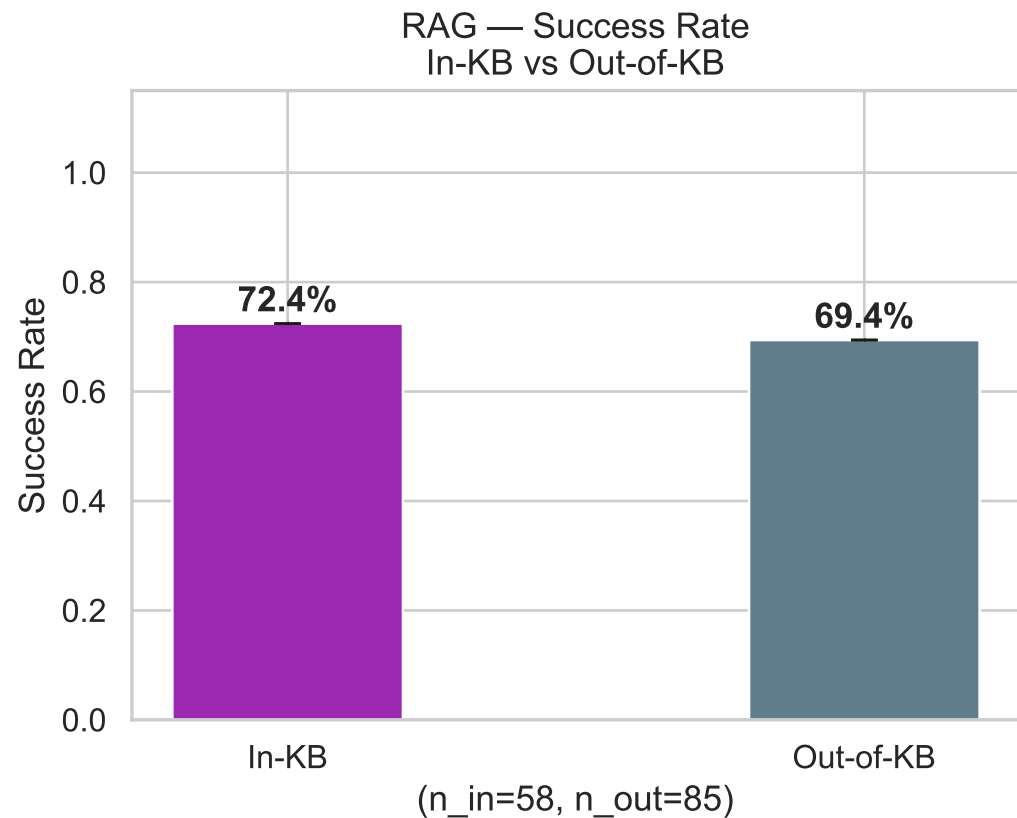
Figure 6 — VLM Score vs Generation Time



**Figure 7 — VLM Score Frequency Distribution**

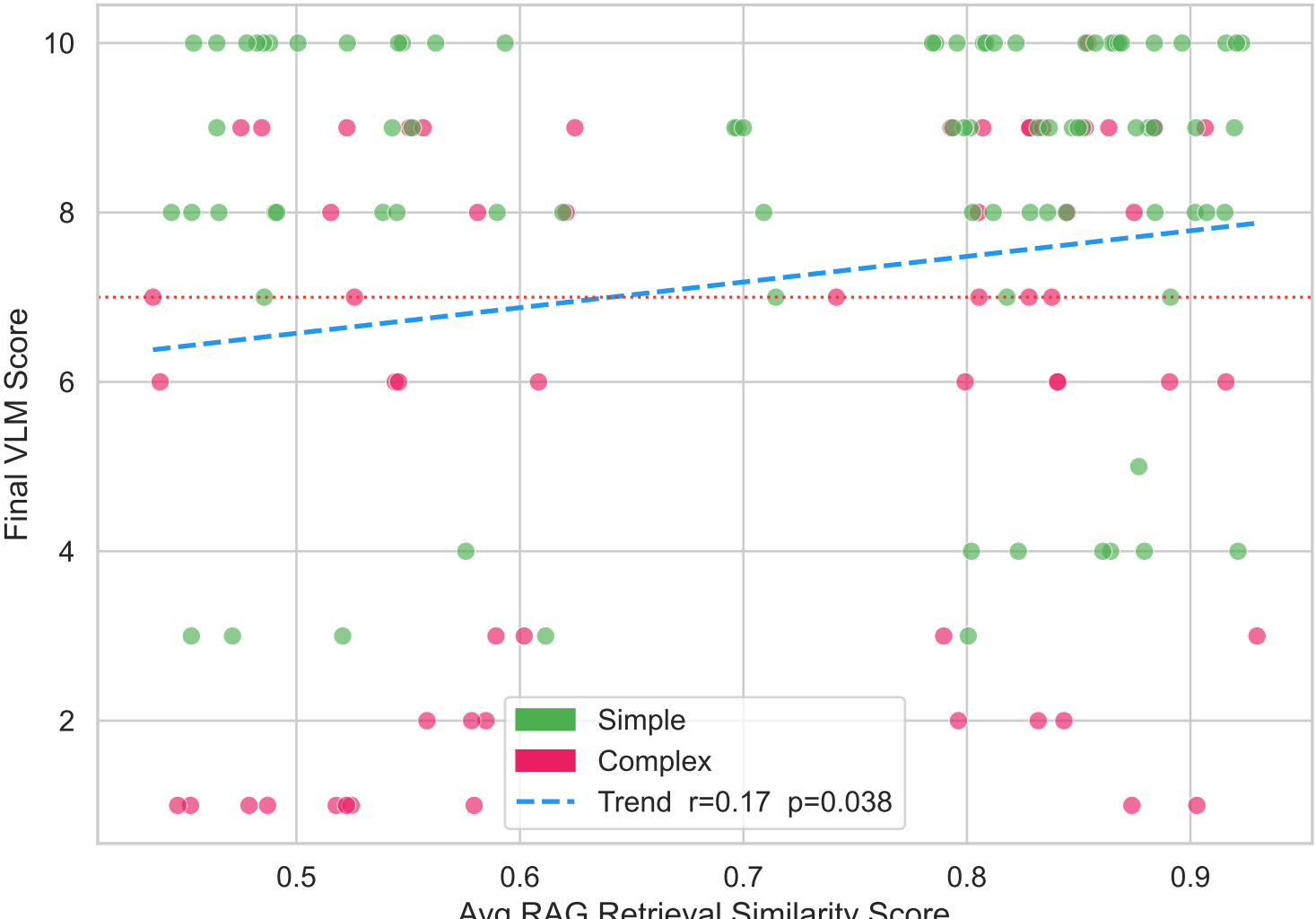


**Figure 8 — Impact of Knowledge-Base Coverage (RAG Only)**

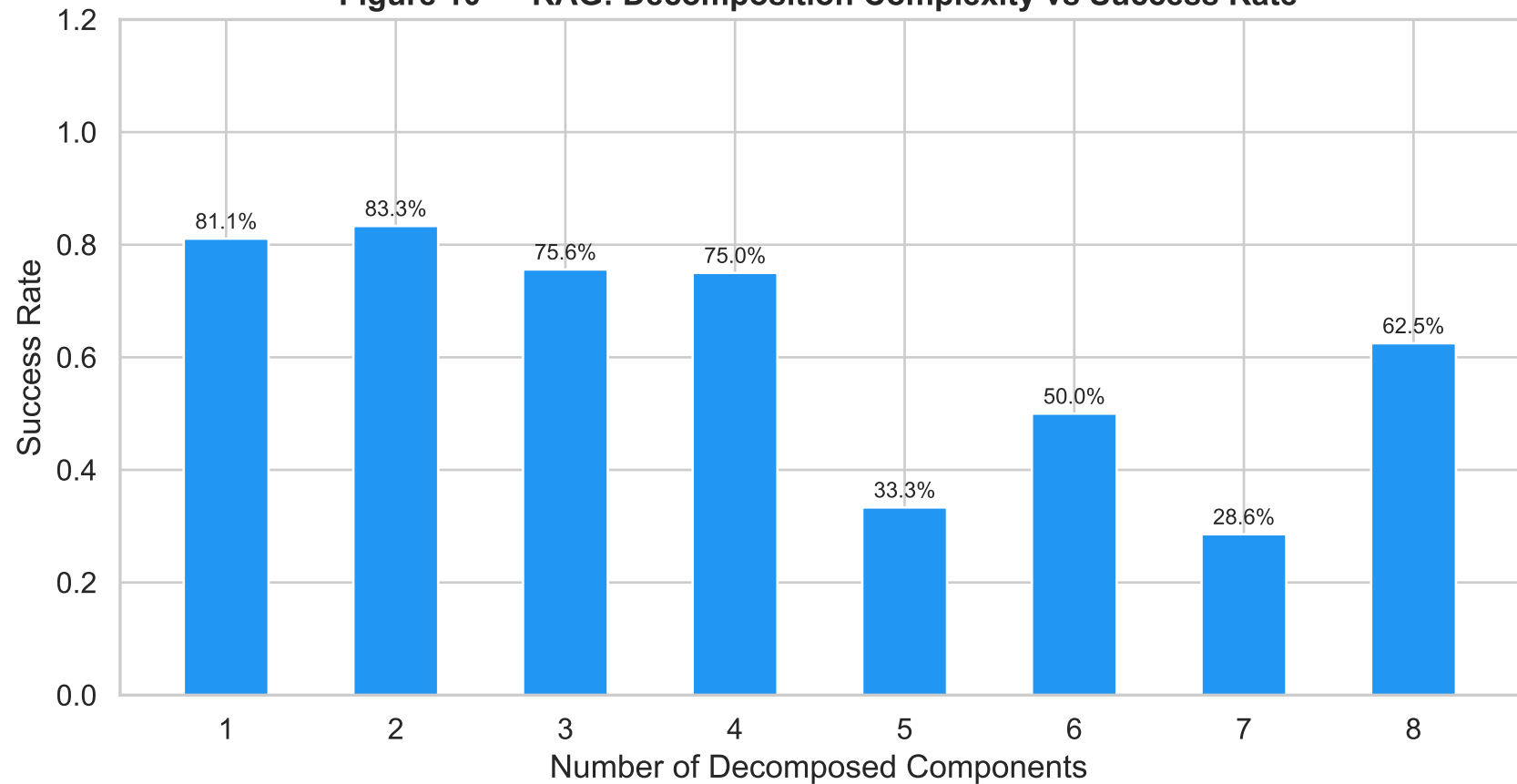




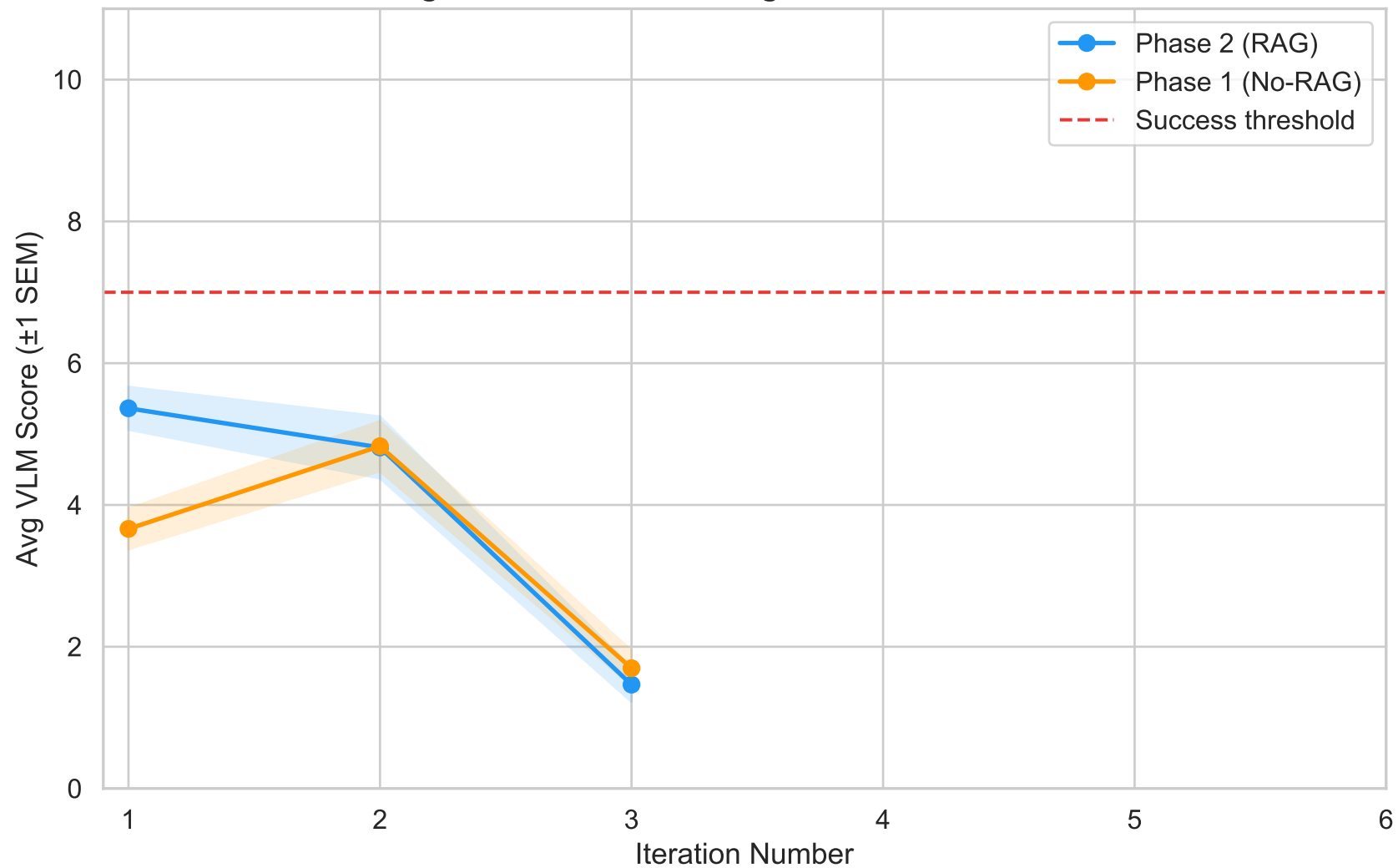
### Figure 9 — Retrieval Similarity vs Generation Quality (RAG)



**Figure 10 — RAG: Decomposition Complexity vs Success Rate**



**Figure 11 — Score Convergence Across Iterations**



**Figure 12 — Success Rate by Code LLM × Eval Provider**

Code LLM

Claude Sonnet

0.72

GPT-5.4

0.58

Gemini 3 Pro Preview

0.88

Gemini 3.1 Pro

0.93

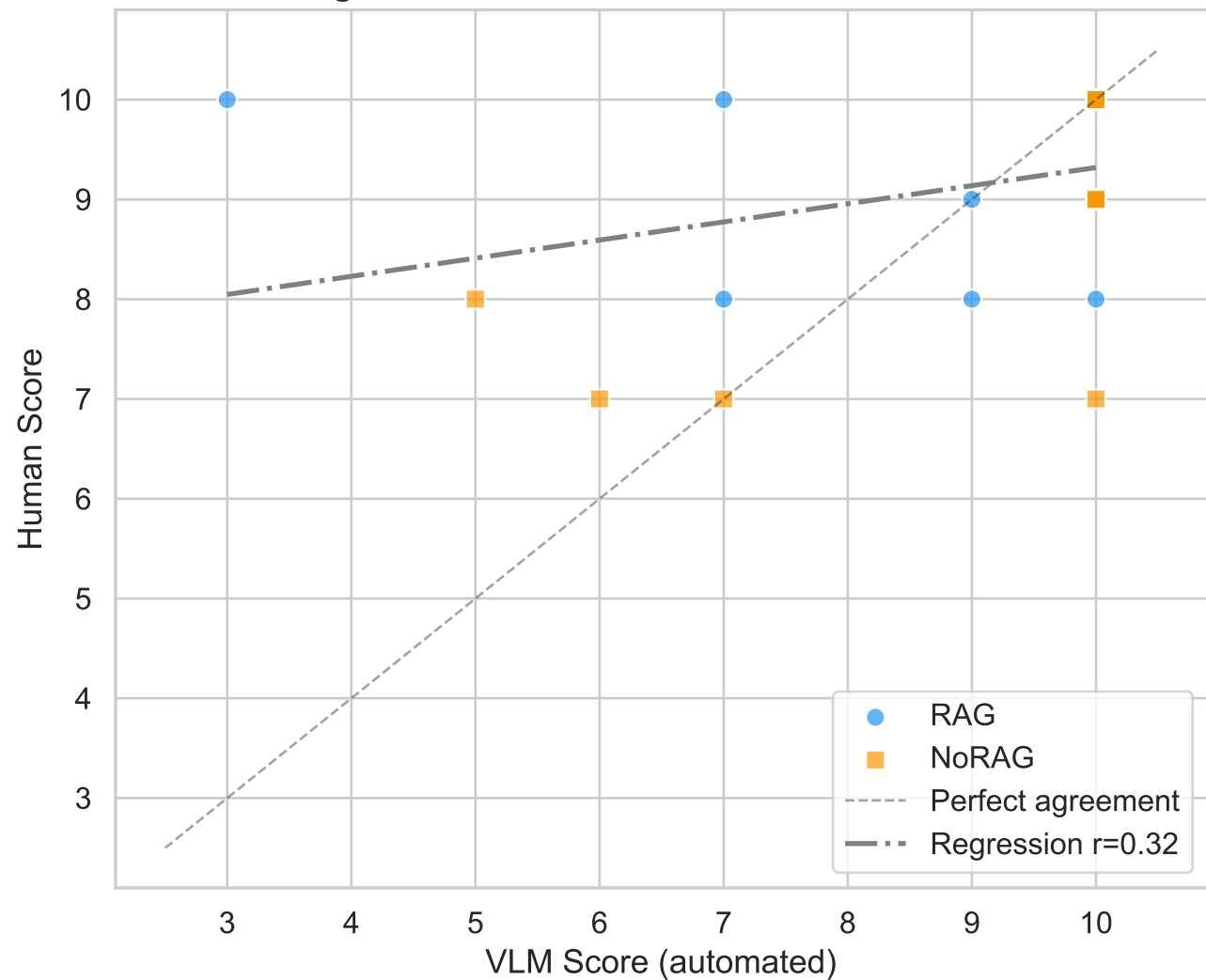
Kimi K2.6

0.24

gemini-3-pro-preview  
Eval Provider



Figure 13 — VLM Score vs Human Evaluation



**Figure 14 — Phase 1 vs Phase 2 Key Metrics Summary**

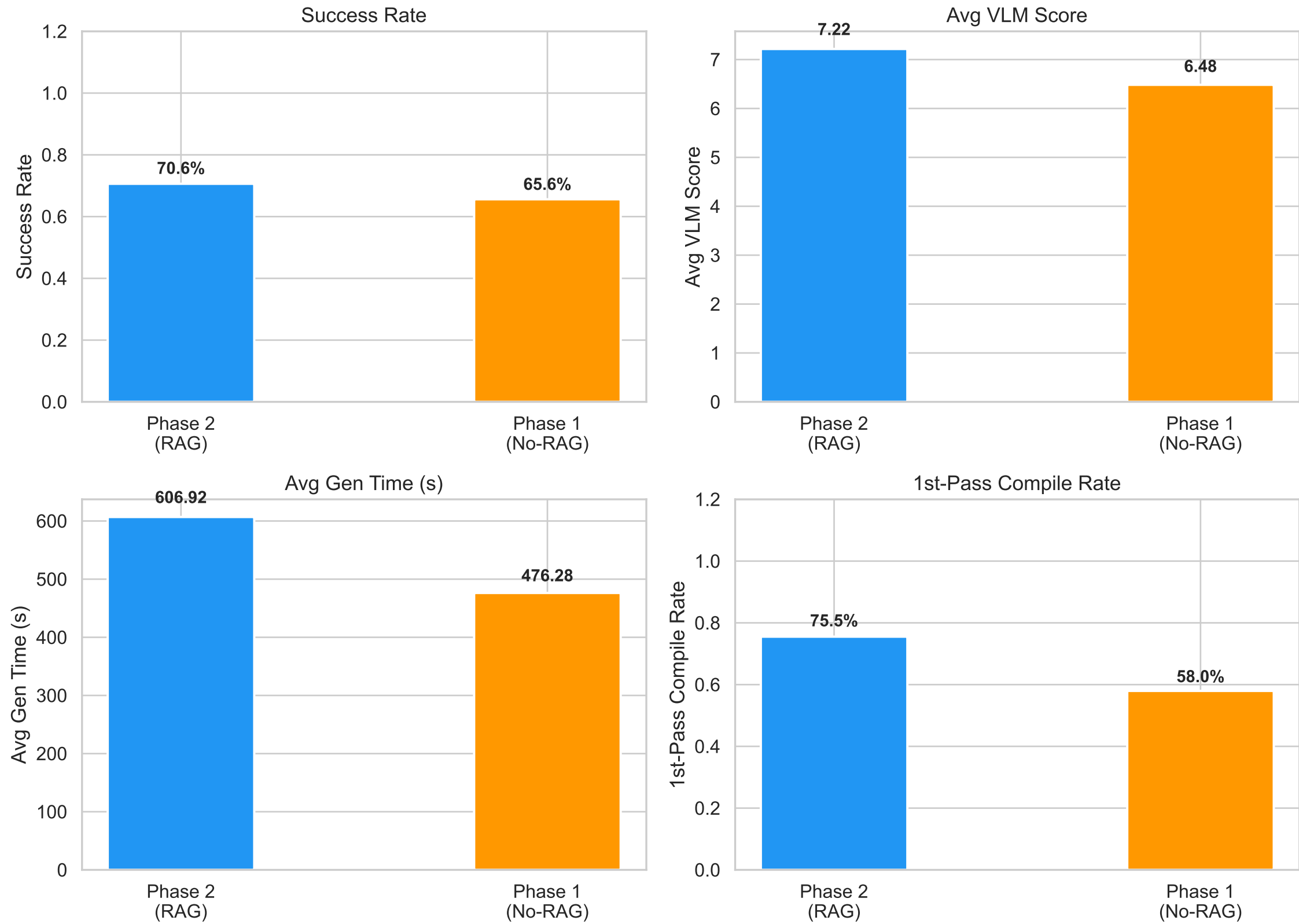
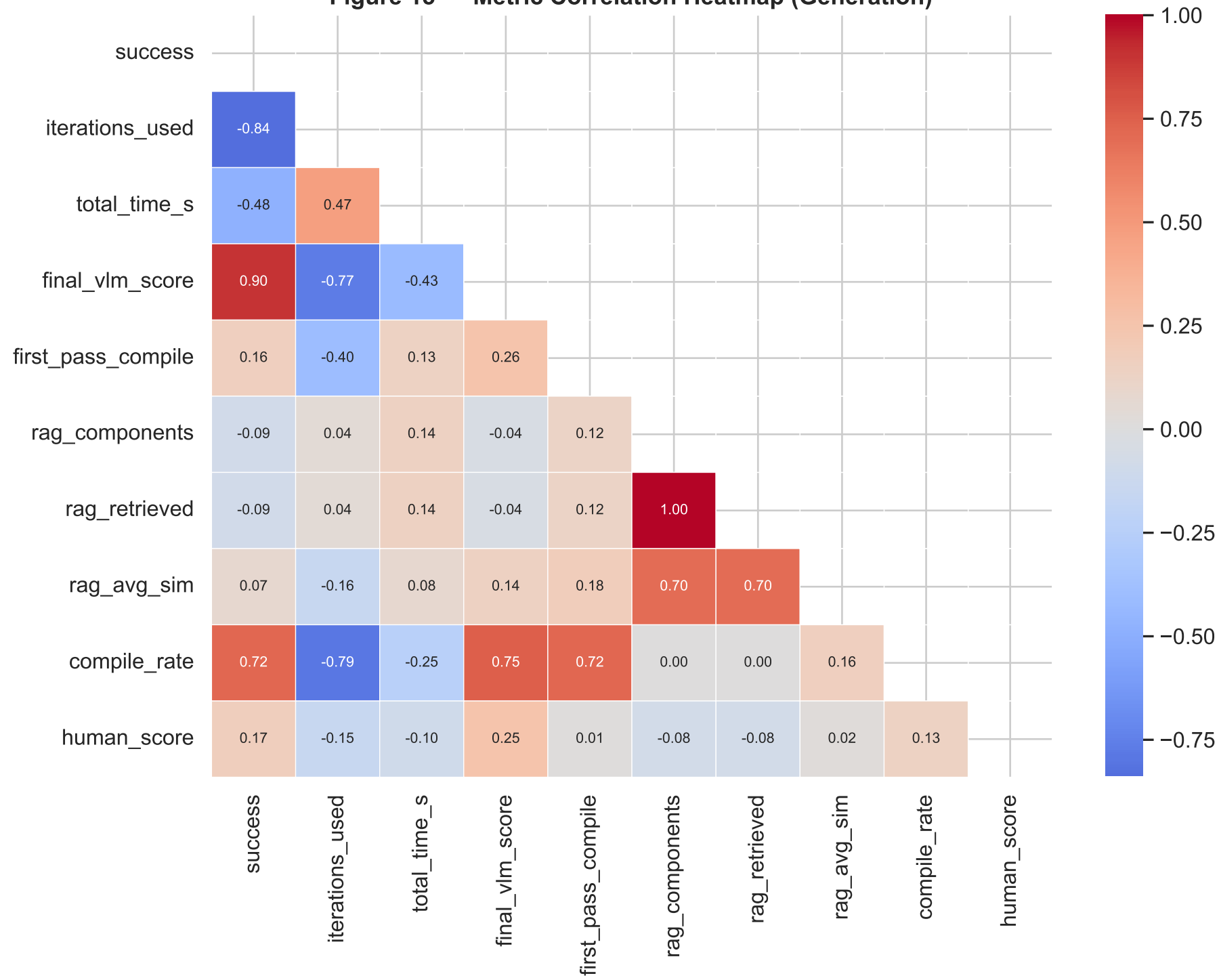
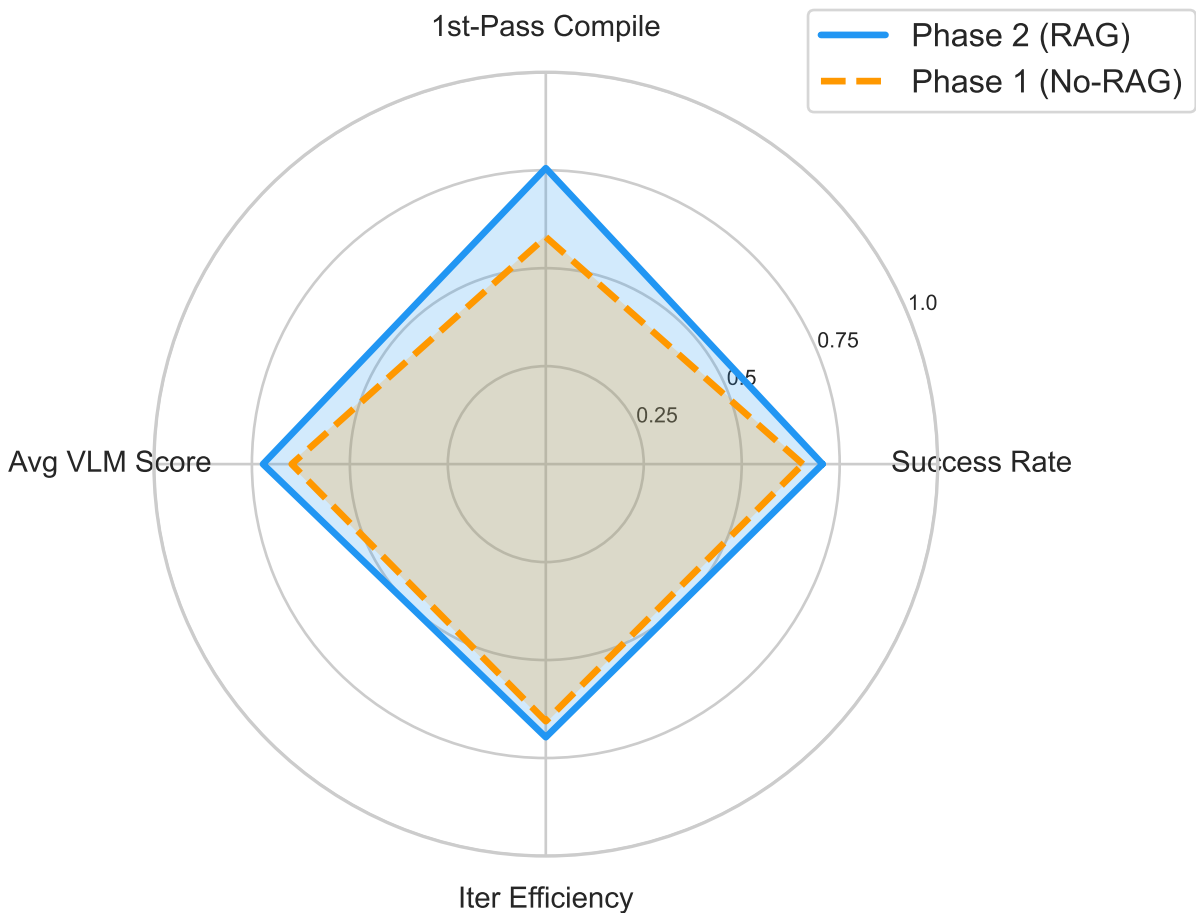


Figure 15 — Metric Correlation Heatmap (Generation)

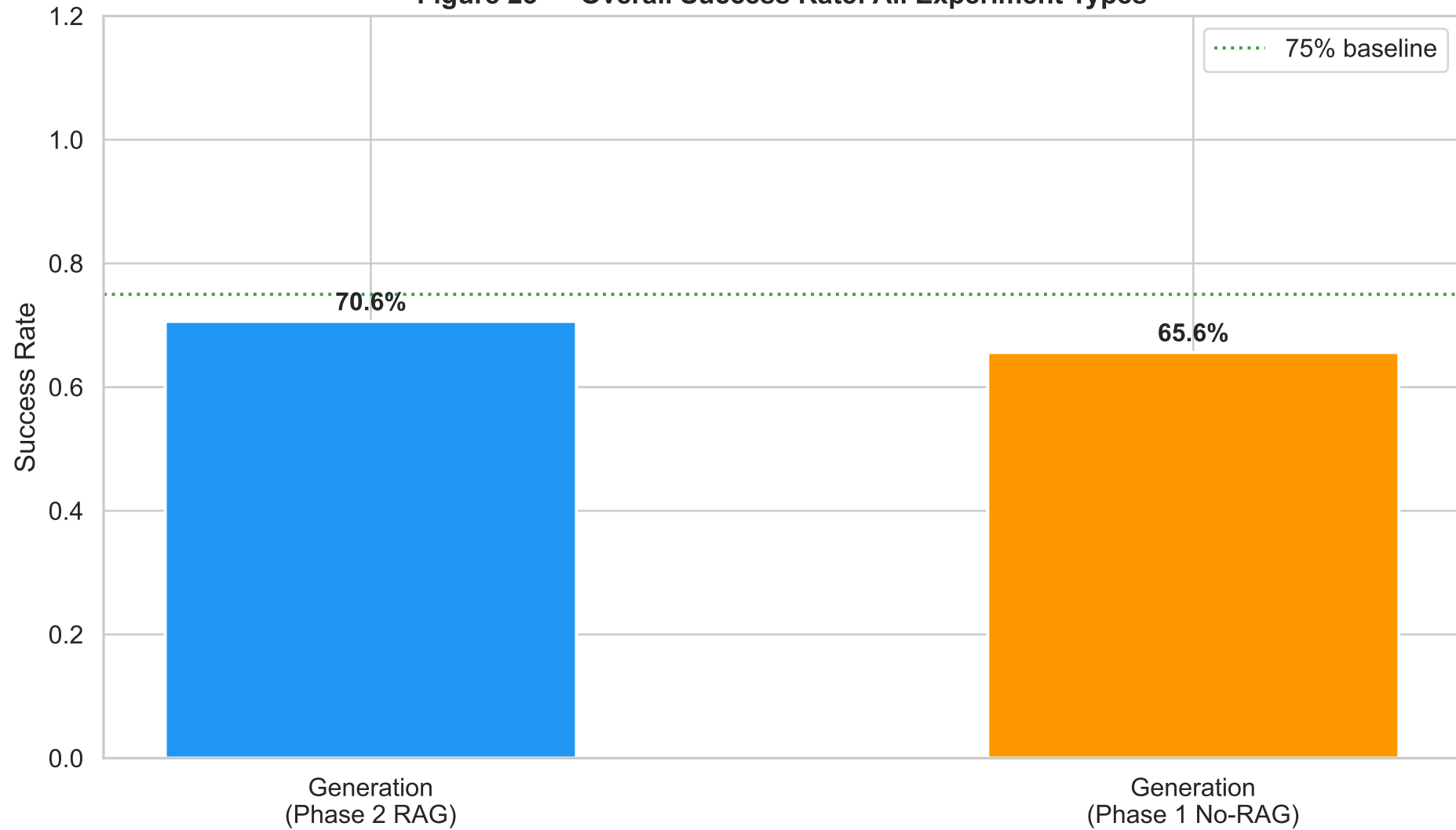


**Figure 16 — Radar: Phase 1 vs Phase 2  
Normalised Metrics**





**Figure 23 — Overall Success Rate: All Experiment Types**



**Figure 24 — Success Rate by LLM Model & Pipeline**

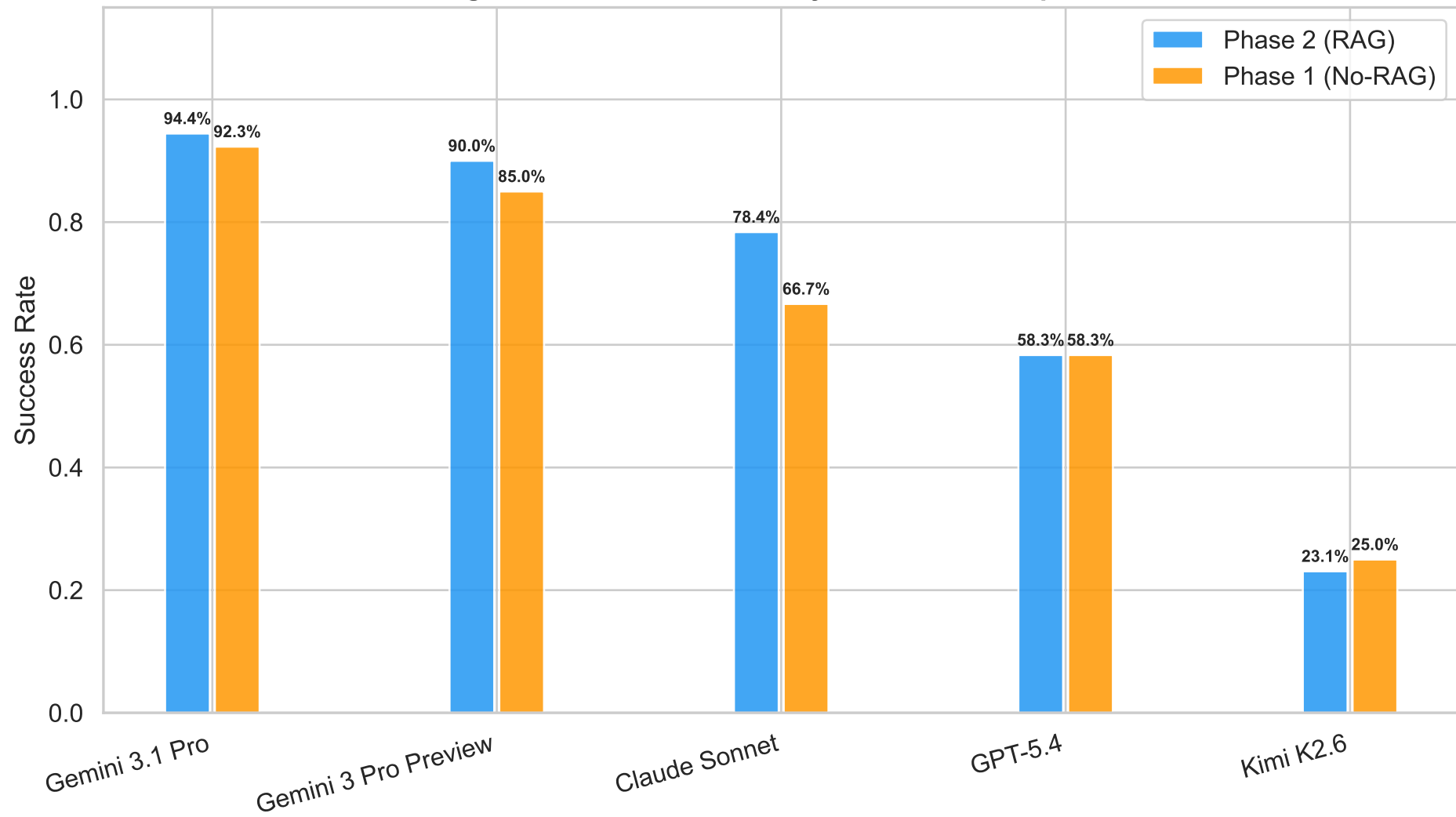
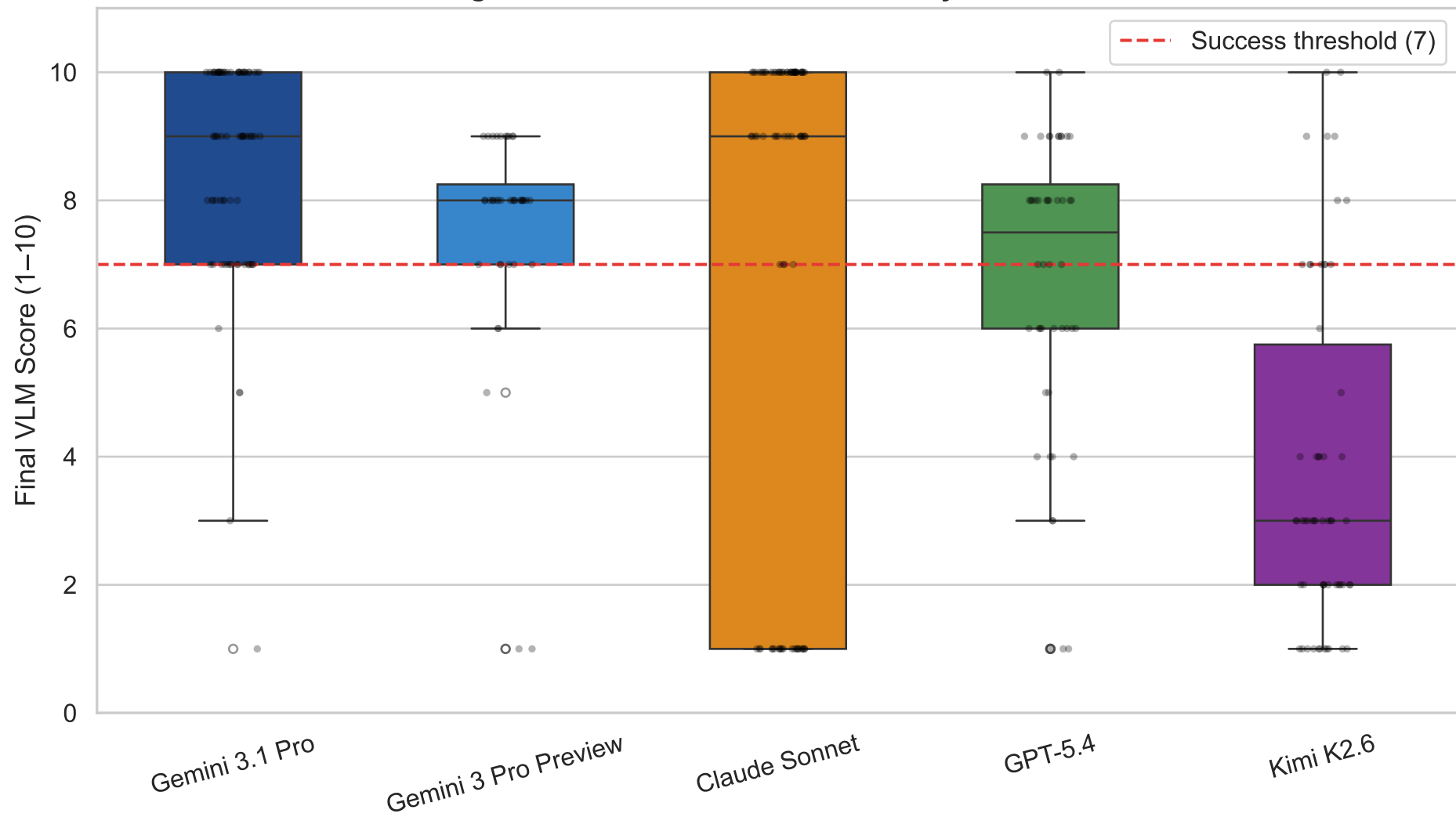
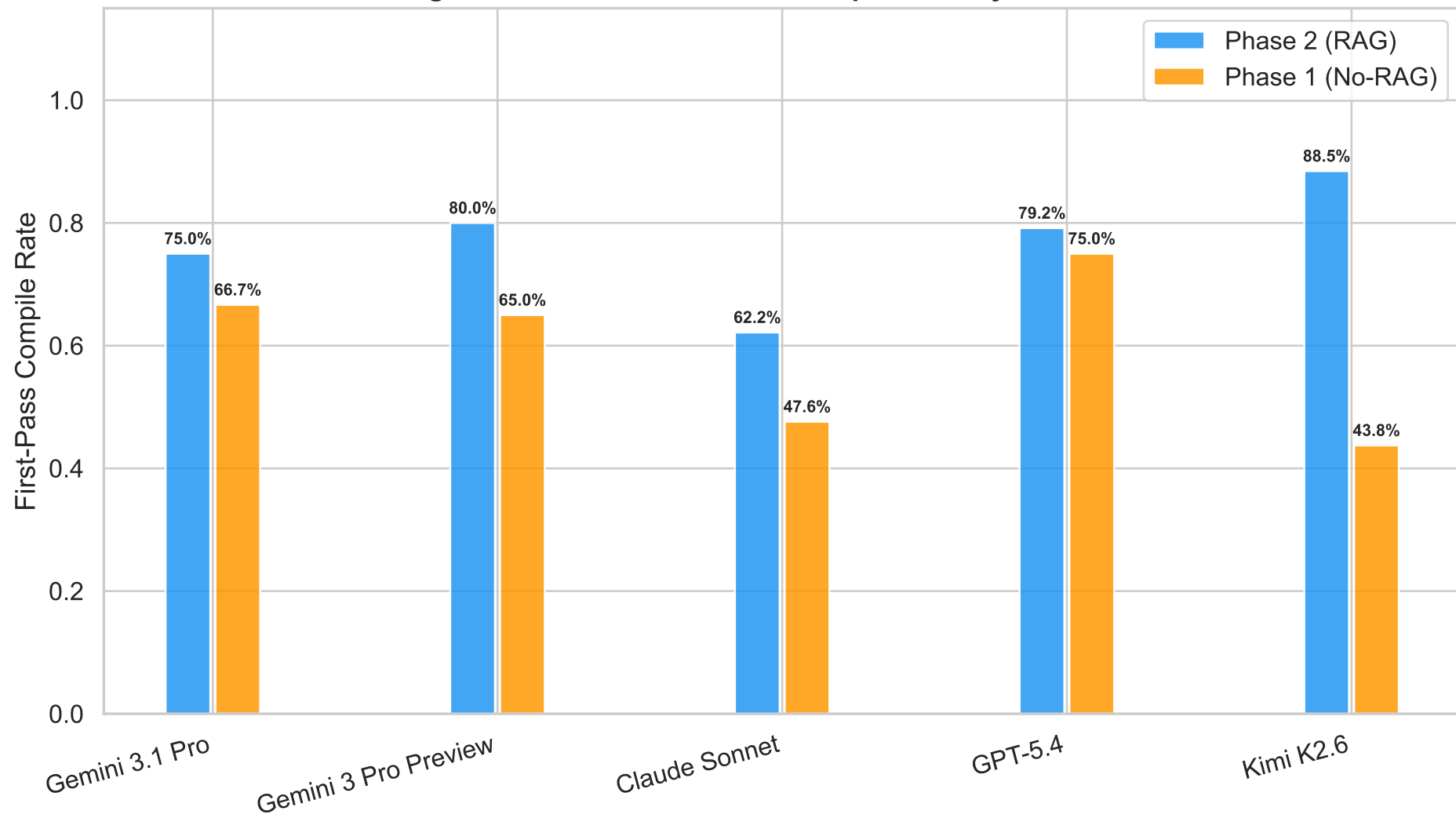


Figure 25 — VLM Score Distribution by LLM Model



**Figure 26 — First-Pass HLSL Compile Rate by LLM Model**



**Figure 27 — Average Iterations to Success by LLM Model**

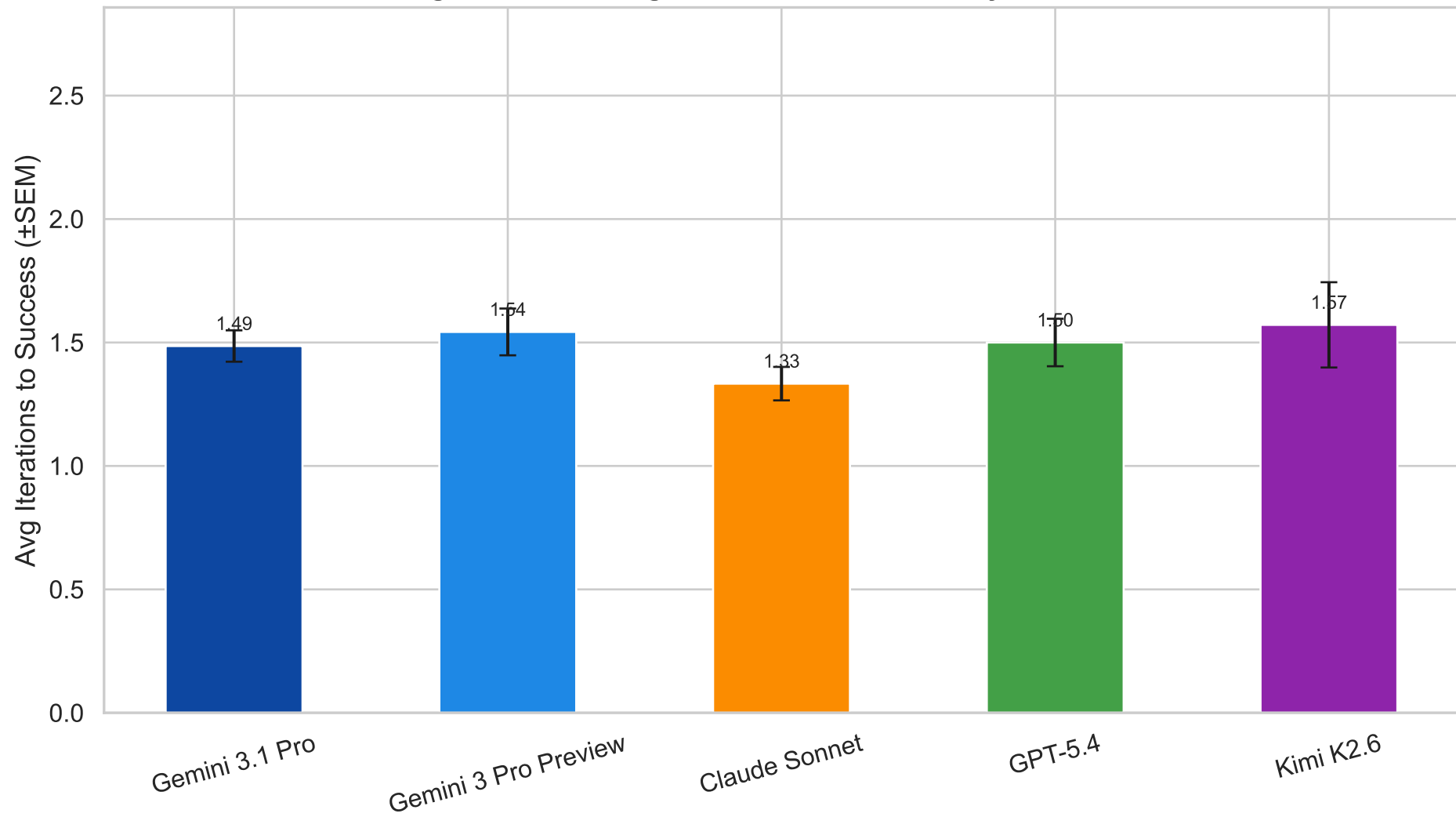
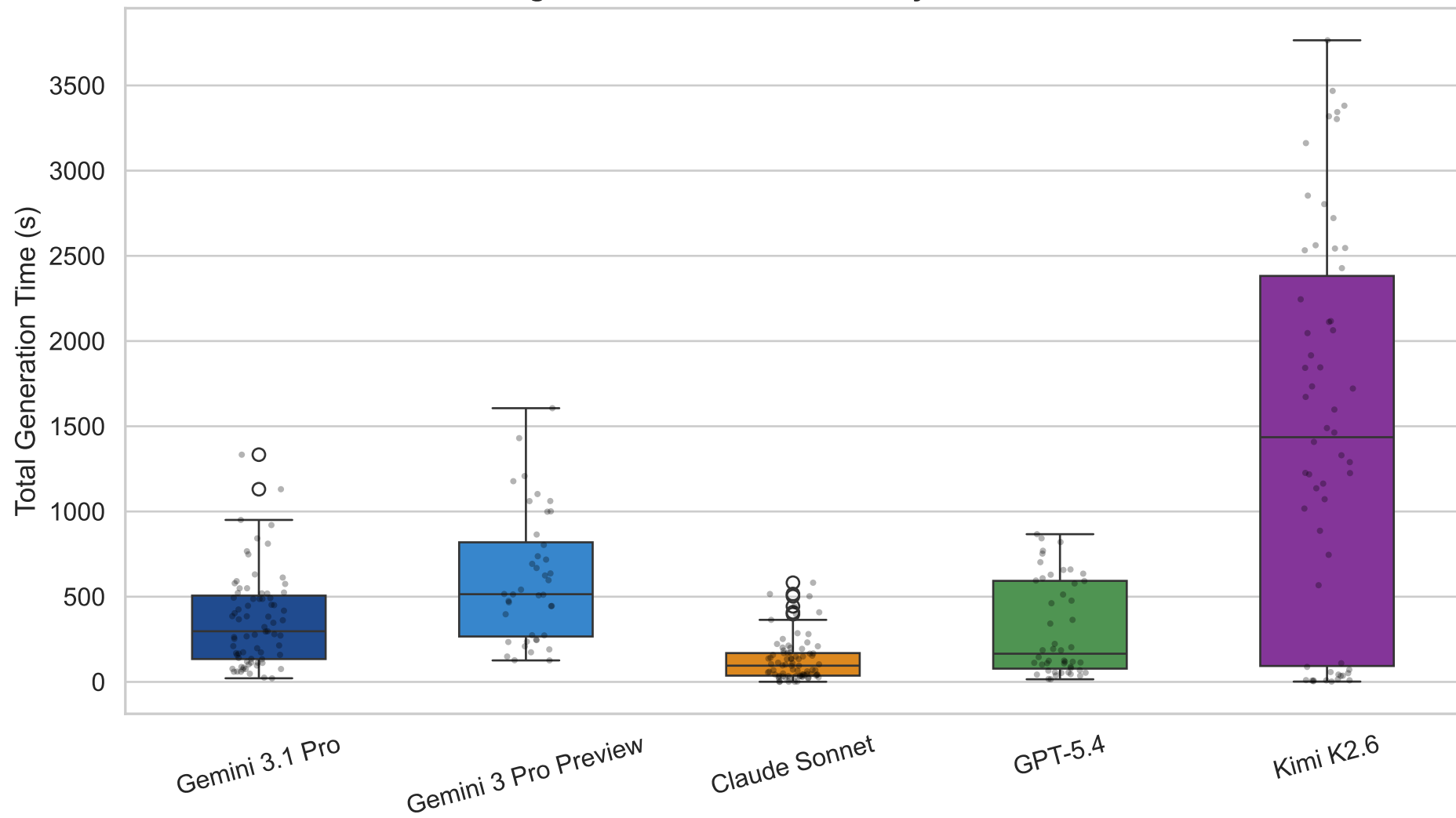


Figure 28 — Generation Time by LLM Model



**Figure 29 — Cost & Token Usage by LLM Model**

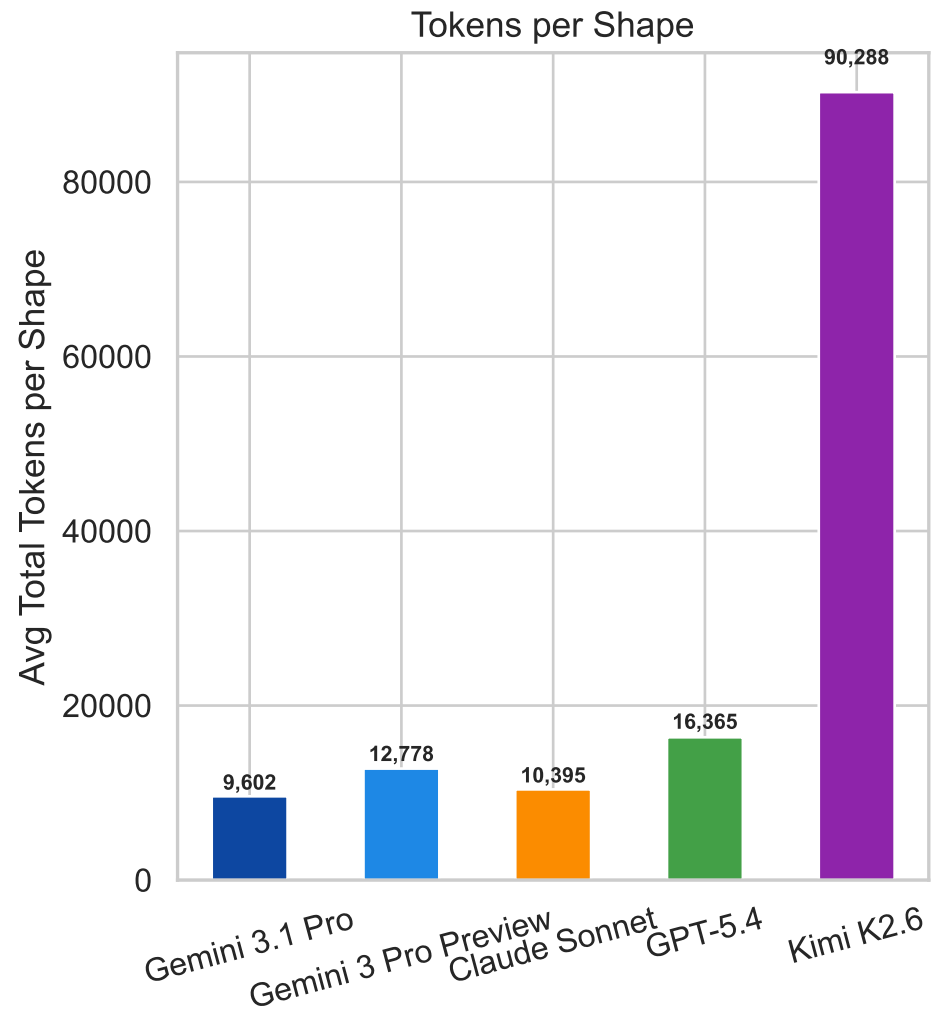
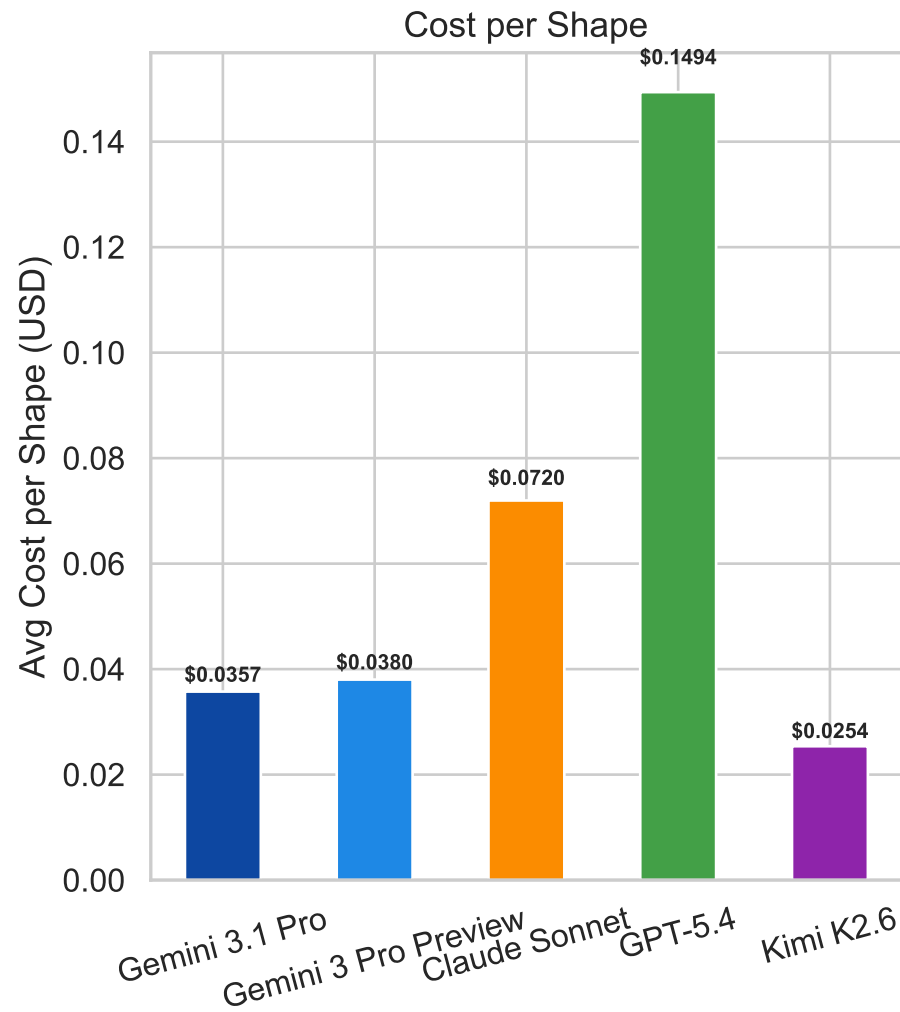


Figure 30 — HLSL Compile Rate per Iteration by LLM Model

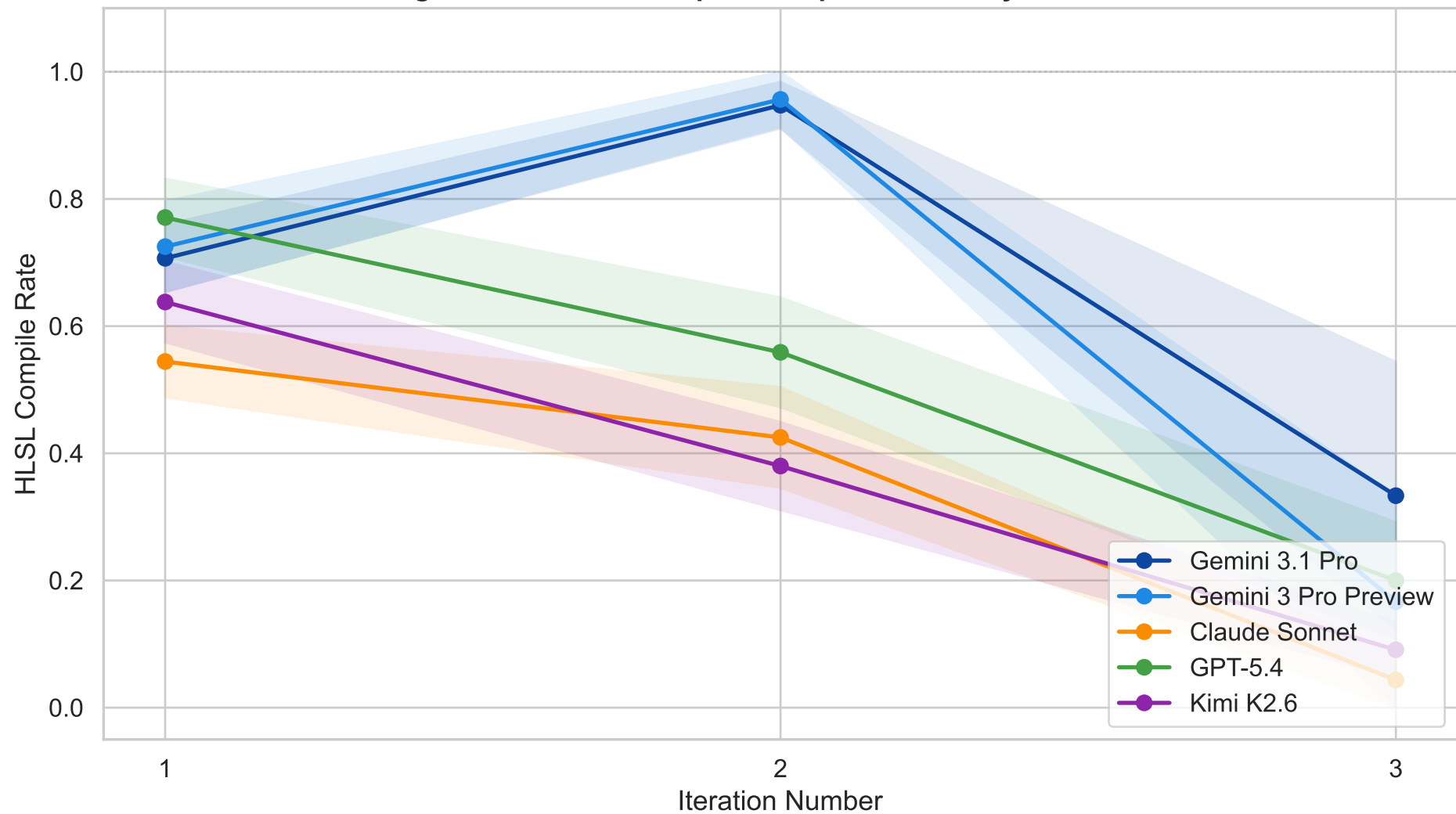
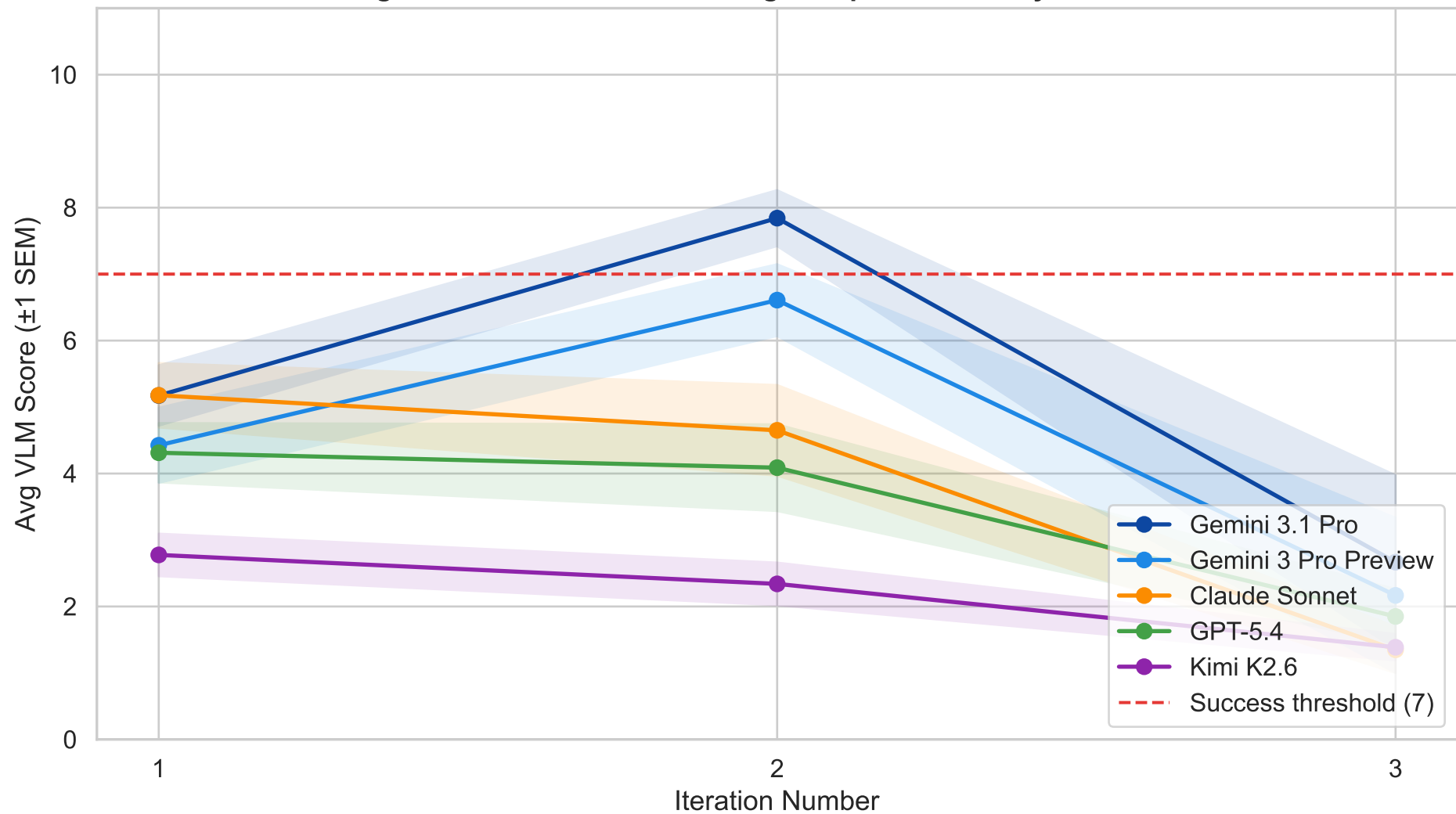




Figure 31 — VLM Score Convergence per Iteration by LLM Model



**Figure 32 — Iteration Recovery Rate by LLM Model**  
(Shapes where Iter 1 failed to compile — did Iter 2 compile?)

